SOFTWARE ENGINEERING & DESIGN PRINCIPLE LAB

Experiment 2: Identification of Problem, Objectives, Scope and Feasibility Analysis

FEASIBILITY REPORT: LEARNING MANAGEMENT SYSTEM

Particular	Details
Experiment No.	2
Project Title	Learning Management System (LMS)
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Semester & Section	5th D
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1. Problem Identification

1.1 Background / Context

Education and training increasingly depend on digital platforms for delivering lessons, managing learners, conducting assessments and monitoring academic performance. A Learning Management System (LMS) is a web-based software system that provides a centralized environment for teachers, students and administrators.

In a traditional academic environment, learning material may be distributed through printed notes, messaging applications, email, physical classrooms and separate examination tools. This makes it difficult to maintain a single, organized record of courses, assignments, attendance, grades and communication.

The proposed LMS will provide one integrated platform through which instructors can create and manage courses, students can access learning resources and submit work, and administrators can supervise users and system activities. The system will be designed primarily for educational institutions, coaching organizations or small training departments.

1.2 Existing Problem

Students and instructors face difficulties when educational activities are managed manually or through multiple unrelated applications. Students may not know where to find the latest study material, assignment instructions, deadlines or grades. Teachers may spend considerable time distributing resources, collecting submissions, checking attendance and preparing performance reports. Administrators may have difficulty maintaining accurate records because data is scattered across registers, spreadsheets, email accounts and messaging groups.

The absence of a centralized system also makes it difficult to monitor student progress. A teacher may not immediately know which students have accessed course content, missed assignments or require additional academic support.

1.3 Existing Method / Current System

- Physical classroom teaching
- Printed notes and handouts
- Email for sharing documents
- Messaging applications for announcements

- Spreadsheets for attendance and marks
- Separate online tools for quizzes or video meetings
- Manual registers for student and course records
- Verbal communication for deadlines and academic updates

1.4 Limitations of the Existing System

1. Learning materials are scattered across paper files, email, messaging groups and personal devices.
2. Students may miss announcements, assignment deadlines or changes in the academic schedule.
3. Teachers must manually maintain attendance, marks, submissions and student records.
4. There is no centralized mechanism for tracking course completion and learner progress.
5. Manual data entry can result in duplicate, incomplete or inaccurate records.
6. Preparing academic reports requires considerable time and effort.
7. Communication between students and teachers is not organized according to courses or topics.
8. It is difficult to provide controlled access to confidential information such as grades and personal records.

2. Problem Statement

Students, instructors and academic administrators face difficulties in managing learning activities because educational content, communication, assignments, attendance and assessment records are handled through disconnected manual and digital methods. The existing approach can cause information loss, missed deadlines, duplication of work, inaccurate records and poor visibility into student performance. Teachers spend time performing administrative tasks instead of concentrating on teaching and feedback, while students lack a single, reliable location for accessing course resources and monitoring their progress. Therefore, a centralized Learning Management System is required to organize academic information, automate routine activities, support communication and provide timely performance reports through a secure web-based platform.

3. Objectives of the Project

S. No.	Objective
1	To develop a centralized web-based platform for managing courses, users and academic activities.
2	To enable administrators to create user accounts, assign roles and manage system access.
3	To allow instructors to create courses and upload study materials in an organized manner.
4	To enable students to enroll in approved courses and access learning resources.
5	To automate assignment creation, submission, deadline management and basic evaluation.
6	To provide online quizzes and assessments with automatic scoring for objective questions.
7	To maintain attendance, marks and progress records in a centralized database.
8	To generate dashboards and reports that help instructors and administrators monitor learner performance.

4. Scope of the Project

4.1 Target Users / Stakeholders

S. No.	User / Stakeholder	Role
1	System Administrator	Manages users, roles, courses, settings and system records.
2	Instructor / Teacher	Creates courses, uploads content, gives assignments, evaluates work and monitors students.
3	Student / Learner	Enrolls in courses, accesses content, submits assignments and attempts assessments.
4	Academic Coordinator	Supervises courses, schedules, attendance and academic reports.
5	Institution Management	Reviews overall usage, course performance and institutional reports.
6	Technical Support Staff	Handles account, access and basic system-support issues.

4.2 Functional Scope

1. User registration and login: secure login, password reset and role-based access.
2. User and role management: add, update, deactivate and search users; assign roles.
3. Course management: create, edit, publish and archive courses; organize modules and lessons.
4. Learning material management: upload notes, documents, links and videos.
5. Enrollment management: enroll students and view enrollment status.
6. Assignment management: create assignments, accept submissions, record marks and feedback.
7. Quiz and assessment management: create objective questions, set time limits and automatically calculate scores.
8. Attendance management: record course attendance and generate attendance reports.
9. Progress tracking: track lesson completion, assignment status and quiz results.
10. Announcements and notifications: publish course updates, deadlines and results.
11. Role-based dashboards for students, instructors and administrators.
12. Reports for attendance, marks, submissions and course participation.

4.3 Data Scope

S. No.	Data / Information	Description
1	User Data	Name, email, contact information, role, login credentials and account status.
2	Course Data	Course title, description, instructor, category, duration and publication status.
3	Learning Content	Notes, documents, video links, lesson descriptions and module information.
4	Enrollment Data	Student-course relationships, enrollment date, approval status and completion state.
5	Assignment Data	Assignment title, instructions, due

		date, maximum marks and attached resources.
6	Submission Data	Student submission file, timestamp, status, marks and feedback.
7	Quiz Data	Questions, options, correct answers, marks, time limits and attempt rules.
8	Result Data	Quiz scores, assignment grades, total marks and instructor comments.
9	Attendance Data	Student, course, date, attendance status and attendance percentage.
10	Notification Data	Announcements, messages, recipients, timestamps and read status.
11	Audit Data	Login activity, important updates and administrative actions.

Sensitive information such as passwords will not be stored in plain text. Passwords will be stored using secure one-way hashing, and access to grades and personal information will be restricted according to user roles.

4.4 Out of Scope

1. Live video conferencing and real-time virtual classrooms.
2. Online payment, subscription and fee-collection facilities.
3. Advanced artificial-intelligence tutoring or automated teaching.
4. Biometric attendance and facial-recognition-based identification.
5. Native Android and iOS applications.
6. Integration with external university examination systems.
7. Automated plagiarism detection through a third-party service.
8. Full support for multilingual content and automatic translation.

5. Technical Feasibility Analysis

Technical feasibility examines whether the proposed LMS can be developed and operated using available hardware, software, technologies and technical skills.

5.1 Proposed Technology Stack

Component	Proposed Technology
Frontend	HTML5, CSS3, JavaScript and NextJS
Backend	Node.js with Express.js
Database	PostgreSQL
Programming Languages	JavaScript / TypeScript, SQL, HTML and CSS
Authentication	Session-based authentication or JWT with role-based access control
API Design	RESTful APIs
File Storage	Local development storage; cloud object storage in deployment
Development Tools / IDE	Visual Studio Code
Version Control	Git and GitHub
Testing Tools	Jest, Supertest and Postman
Deployment	Linux server or cloud platform
Other Technologies	HTTPS, database migrations, responsive design and email

	notification service

NextJS is suitable for creating reusable interface components and dashboards. Node.js and Express.js can provide APIs for authentication, courses, assessments and reports. PostgreSQL is appropriate because the LMS contains strongly related entities such as users, courses, enrollments, assignments, submissions and results.

The application should follow secure software-development practices. The OWASP Top 10 identifies major web-application security risks such as broken access control, security misconfiguration, cryptographic failures, injection and authentication failures.

The user interface should consider accessibility. WCAG 2.2 provides recommendations for making web content accessible to people with visual, auditory, physical, cognitive and learning disabilities.

5.2 Hardware Requirements

Hardware	Requirement
Processor	Dual-core 64-bit processor or better
RAM	Minimum 8 GB; 16 GB recommended
Storage	At least 20 GB free SSD storage
Display	1366 × 768 resolution or higher
Network	Stable internet connection for package installation, collaboration and testing

Deployment Server

Hardware	Requirement
Processor	2 virtual CPU cores for a small pilot
RAM	4 GB minimum; 8 GB recommended
Storage	50 GB SSD initially, expandable as files increase
Backup Storage	Separate location for database and uploaded-file backups
Network	Reliable internet connection with HTTPS support
Other Hardware	Optional monitoring and backup service

5.3 Software Requirements

Software	Version / Requirement
Operating System	Windows 10/11, Ubuntu Linux or another supported 64-bit OS
IDE / Editor	Visual Studio Code
Runtime	Node.js LTS
Database	PostgreSQL
Frontend Framework	NextJS
Backend Framework	Express.js
Browser	Google Chrome, Mozilla Firefox, Microsoft Edge or equivalent
API Testing	Postman
Version Control	Git
Database Tool	pgAdmin or command-line PostgreSQL tools
Other Software	npm, environment-variable manager and backup utilities

5.4 Technical Skills Required

- HTML, CSS and responsive web design.
- JavaScript or TypeScript programming.
- NextJS component development.
- Node.js and Express.js backend development.
- REST API design and integration.
- Relational database design and SQL.
- Authentication and role-based authorization.
- File upload and file-validation techniques.
- Git and collaborative software development.
- Unit, integration and user-acceptance testing.
- Basic deployment, logging and backup management.
- Secure coding practices, including input validation and protection against injection.

5.5 Technical Risks and Controls

Technical Risk	Possible Effect	Proposed Control
Unauthorized access	Student or grade data may be exposed.	Apply role-based access control and test authorization rules.
Weak passwords	Accounts may be compromised.	Use password hashing and password policies.
Malicious file uploads	Server or users may be exposed to harmful files.	Restrict file types, limit file sizes, scan files and store uploads safely.
Data loss	Academic records may be lost.	Perform scheduled database and file backups.
Poor performance	Users may experience delays during peak periods.	Use pagination, indexing, caching and performance testing.
Incorrect assessment results	Students may receive inaccurate marks.	Validate scoring logic and test multiple cases.
Service unavailability	Users may be unable to access courses.	Monitor uptime, maintain backups and prepare recovery procedures.
Accessibility problems	Some users may be unable to use the system effectively.	Follow WCAG 2.2 principles and perform keyboard and screen-reader checks.

5.6 Technical Feasibility Assessment

The proposed LMS is technically feasible. The required development tools are freely available or have accessible educational versions. The system can be developed using common web technologies, and the expected modules are appropriate for a student software-engineering project.

The project should be implemented incrementally. The first development phase should focus on authentication, role management, course management, learning materials and assignments. Assessments, attendance, reports and notifications can then be added and tested.

The main technical challenges will involve authorization, file management, data security, assessment correctness and responsive user-interface design. These challenges are manageable if the team follows a modular architecture, uses version control, conducts regular testing and limits the initial scope.

Conclusion

It is Technically feasible.

Justification: The project can be developed with available computers, open-source technologies, a relational database and standard web-development skills. The system does not require highly specialized hardware or experimental technology. The scope can be completed as a web application if advanced features such as live video, payment processing and artificial-intelligence tutoring are excluded from the first release.

6. Operational Feasibility Analysis

Operational feasibility determines whether the proposed LMS can be used effectively by its intended users and integrated into an educational environment.

6.1 User Acceptance

The system is likely to be accepted because students, teachers and administrators commonly require a single location for academic information. Students benefit from organized resources, online submissions and progress visibility. Instructors benefit from reduced manual work and easier tracking of student performance. Administrators benefit from centralized records and structured reporting.

User acceptance will depend on simplicity, reliability, mobile responsiveness and the availability of support. The system should be introduced through a pilot course before wider institutional deployment.

6.2 Ease of Use

- Simple navigation with separate dashboards for each role.
- Clear labels for courses, assignments, results and announcements.
- Consistent buttons and page layouts.
- Search and filtering for courses and submissions.
- Responsive design for laptops, tablets and mobile browsers.
- Clear error messages and confirmation messages.
- Visible deadlines and submission status.
- Accessible forms and keyboard navigation.
- Minimal number of steps for common actions.

For example, a student should be able to open the dashboard, select a course, view an assignment, upload a file and receive a submission confirmation without navigating through unrelated pages.

6.3 Existing Workflow

Existing Activity	Proposed LMS Workflow
Teacher distributes notes manually	Teacher uploads material to the relevant course module.
Student searches messages for files	Student accesses all approved content from the course page.
Assignment submitted physically or by email	Student submits through the assignment page.
Teacher records marks in a spreadsheet	Teacher enters marks and feedback directly in the system.
Attendance maintained in a register	Attendance is recorded against each course and date.
Announcements sent through multiple channels	Course announcements are displayed in one location.
Reports prepared manually	Authorized users generate filtered reports.

The LMS will support the existing classroom process rather than forcing an immediate change to every institutional practice. Paper-based activities can continue during the transition period.

6.4 Training Requirements

- Students may need a short orientation covering login, course access, material viewing, assignment submission and quiz attempts.
- Instructors may need training on course creation, material upload, assignment evaluation, attendance and reporting.
- Administrators may need training on user management, role assignment, course approval and backups.
- A user manual, frequently asked questions and short demonstration videos should be provided.
- Helpdesk support should be available during the initial implementation period.

A pilot implementation with a small number of users can identify usability problems before full deployment.

6.5 Expected Operational Benefits

1. Centralized access to course content and academic information.
2. Reduced paperwork and repetitive administrative work.
3. Faster distribution of learning material and announcements.
4. More organized assignment submission and evaluation.
5. Improved visibility into attendance, marks and learner progress.
6. Better communication between instructors and students.
7. Easier preparation of academic reports.
8. Increased flexibility because students can access approved resources outside classroom hours.
10. Improved record accuracy through validation and centralized storage.
11. More consistent academic processes across courses.

6.6 Operational Risks and Controls

Operational Risk	Effect	Proposed Control
Users resist changing existing practices	Low adoption.	Conduct demonstrations and introduce the system gradually.
Users forget passwords	Increased support requests.	Provide password reset and account-recovery features.
Poor internet connectivity	Difficulty accessing content.	Optimize pages, support downloadable resources and permit reasonable submission retries.
Teachers fail to update content	Students receive outdated information.	Provide content status, reminders and course-administration guidelines.
Incorrect data entry	Inaccurate attendance or grades.	Use validation, confirmation screens and controlled editing.
Lack of technical support	Delayed resolution of problems.	Provide documentation and assign support responsibility.

Operational Feasibility

Justification: The LMS fits naturally into academic activities and addresses existing problems faced by students, instructors and administrators. Users can access the platform through standard web browsers, and the main workflows are straightforward. With basic training, documentation, technical support and phased implementation, the proposed system can be adopted successfully.

7. Overall Feasibility Summary

Feasibility Area	Status	Justification
Technical Feasibility	Feasible	Required hardware, software, development tools and technical skills are available. The system can be built using established web technologies.
Operational Feasibility	Feasible	The LMS supports existing academic workflows, reduces manual effort and provides useful dashboards and reports.
Overall Project Feasibility	Feasible	The proposed first version has a manageable scope and can be developed incrementally by a student software team.

Overall Conclusion

Based on the technical and operational analysis, the proposed Learning Management System is feasible for development. The system can be implemented with commonly available hardware, open-source technologies and standard web-development skills. It provides clear benefits by centralizing course information, automating assignments and assessments, maintaining academic records and improving communication.

The project should begin with a minimum viable product containing authentication, role management, course management, learning materials, assignments and basic reports. More complex features should be postponed until the core system is stable. Proper attention must be given to security, privacy, backups, accessibility and user training.

8. Expected Outcome

After completion, the LMS is expected to provide a secure and centralized platform for academic management. Administrators will be able to manage users and courses, instructors will be able to deliver content and evaluate students, and students will be able to access resources, submit assignments and monitor their progress.

- Reduce dependence on paper records and scattered communication channels.
- Provide organized course and learning-material management.
- Improve the submission and evaluation process.
- Maintain attendance, marks and progress records.
- Generate useful academic reports.
- Support role-based and secure access.
- Make learning information available from any supported web browser.
- Provide a foundation for future mobile, analytics and integration features.

9. Future Enhancements

1.Native Android and iOS applications.
2. Live video classes and virtual meeting integration.
3. Discussion forums and real-time chat.
4. Artificial-intelligence-based learning recommendations.
5. Automated plagiarism detection.

6. Advanced analytics and predictive student-performance reports.
7. Online fee payment and subscription management.
8. Digital certificates and course-completion badges.
9. Integration with Google Classroom, Microsoft Teams or institutional information systems.
10. Biometric or QR-code-based attendance.
11. Multilingual user interfaces and translated course content.
12. Offline access and synchronization for users with limited internet connectivity.
13. Multi-factor authentication and single sign-on.
14. SCORM or xAPI support for reusable e-learning content.
15. Parent or guardian access to approved student-progress information.

References

- W3C, Web Content Accessibility Guidelines (WCAG) 2.2. https://www.w3.org/TR/WCAG22/
- OWASP, OWASP Top 10:2025. https://owasp.org/Top10/2025/
- OWASP, OWASP Top Ten Web Application Security Risks. https://owasp.org/www-project-top-ten/